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June 27, 2025

JN 25173

Derek Richardson
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Subject: **Transmittal Letter – Geotechnical Engineering Study**
Proposed Residence Additions
8560 + 8562 Sand Point Way Northeast
Seattle, Washington

Greetings:

Attached to this transmittal letter is our geotechnical engineering report for the proposed residence additions to be constructed in Seattle, Washington. The scope of our services consisted of exploring site surface and subsurface conditions and then developing this report to provide recommendations for general earthwork and design considerations for foundations, retaining walls, subsurface drainage, and temporary excavations. This work was authorized by your acceptance of our proposal, dated May 15, 2025.

The attached report contains a discussion of the study and our recommendations. Please contact us if there are any questions regarding this report, or for further assistance during the design and construction phases of this project.

Respectfully submitted,

GEOTECH CONSULTANTS, INC.

James H. Strange, Jr., PE
Principal Engineer

JHS:kg

GEOTECHNICAL ENGINEERING STUDY
Proposed Residence Additions
8560 + 8562 Sand Point Way Northeast
Seattle, Washington

This report presents the findings and recommendations of our geotechnical engineering study for the site of the proposed residence addition project to be located in Seattle.

Development of the property is in the planning stage, and detailed plans were not available to us at the time of this study. We were provided with a preliminary site plan. Based on this plan, we understand that a new house will be constructed in a similar footprint to the existing house and that the new house will reuse the existing foundations of the existing house. An attached garage addition is envisioned for the western side of the southern portion of the house, and an enlarged driveway is also planned.

If the scope of the project changes from what we have described above, we should be provided with revised plans in order to determine if modifications to the recommendations and conclusions of this report are warranted.

SITE CONDITIONS

SURFACE

The Vicinity Map, Plate 1, illustrates the general location of the site in the Sand Point neighborhood of Seattle. The ~18,000 square-foot generally rectangular shaped site has about 65 feet of frontage along Sand Point Way Northeast to the west and about 265 feet of landed depth in the east-west direction. The site is bordered to the east by Lake Washington and the west by the street. Residential parcels are located north and south of the site.

The site is currently developed with a single-story, single-family residence which is situated on the eastern side of the property. The residence has a full basement that daylights to the east. We did note some foundation cracks in the south side of the foundation where the structure begins to daylight. The contractor exposed the foundations in two locations on the eastern side of the house, and the footings were 5.5 inches outside the foundation stem wall, 6 inches thick, and bottomed about 21 to 26 inches below the existing grade. The house is accessed via an asphalt driveway that enters the site at its northeastern corner and curves around to the south and east. A detached garage and carport are located north and west of the turnaround at the bottom of the driveway. An open drainage ditch is located near the northern property line. Portions of this ditch are channeled into underground culverts.

The site slopes from west to east, from Sandpoint Way Northeast towards the water. The western edge of the subject site drops down from the street edge at approximately 30-50 percent and a height of just about 10 feet. Rockeries are located on both the upslope and downslope sides of the existing driveway, and the remainder of the proposed development area is relatively flat. The area surrounding the driveway is mulched and landscaped and scattered with mature evergreen trees. No indications of slope instability were observed during our site visits. The existing house has a full basement that daylights to the lower grass yard to the east.

The site is located within a mapped Steep Slope area, Potential Slide Area, and a Liquefaction Hazard Area on the City of Seattle DCLU Critical Areas map.

The adjacent house to the south is about 5 feet from the common property line and is supported on a combination of conventional footings (in the western deep basement area) and pipe piles on the eastern end. And the northern adjacent property is developed with a conventionally founded residence with a full basement.

SUBSURFACE

The subsurface conditions were explored by drilling three test borings at the approximate locations shown on the Site Exploration Plan, Plate 2. We also drilled two borings for an old study of the western lot for a planned separate SFR in 2001. Our exploration program was based on the proposed construction, anticipated subsurface conditions and those encountered during exploration, and the scope of work outlined in our proposal.

The current test borings were drilled on June 4, 2025 using a track-mounted, hollow-stem auger drill. Samples were taken at approximately 2.5- to 5-foot intervals with a standard penetration sampler. This split-spoon sampler, which has a 2-inch outside diameter, is driven into the soil with a 140-pound hammer falling 30 inches. The number of blows required to advance the sampler a given distance is an indication of the soil density or consistency. A geotechnical engineer from our staff observed the drilling process, logged the test borings, and obtained representative samples of the soil encountered. The Test Boring Logs are attached as Plates 3 – 5. Previous (2001) borings are also attached.

Soil Conditions

Boring B-1 (located in the proposed garage area) revealed 8 feet of loose to very loose very silty sand overlying medium-dense silty sands that became dense at 15 feet. Borings B-2 (in the existing courtyard) and B-3 (in the eastern yard) also encountered loose silty sands and sandy silts that became medium dense at about 15 feet and dense at 25 feet.

In our previous borings from 2001:

Boring 1, which was located on the southern side of the proposed residence, encountered approximately 2 feet of loose fill and 3 feet of loose, wet silt with sand, gravel, and abundant organics overlying medium-dense to dense, silty sand. The silty sand became fine-grained and dense approximately 15 feet below existing grade and was encountered to the maximum explored depth of 33 feet below existing grade. Boring 2, which was located on the northern side of the proposed residence, encountered approximately 3 feet of loose topsoil and fill soils and 11 feet of loose, silty sand overlying the medium-dense to dense, silty sand. Hard, slightly sandy silt was encountered at 26 feet below existing grade. These soil conditions are consistent with those found on the adjacent sites.

Based on the observed soil samples, and the recorded SPT blow counts, it is apparent that the underlying very stiff and harder, as well as dense and denser native soils have been glacially compressed.

No obstructions were revealed by our explorations. However, debris, buried utilities, and old foundation and slab elements are commonly encountered on sites that have had previous development.

Groundwater Conditions

Groundwater seepage was observed at a depth of 3 to 5.5 feet. The test borings were left open for only a short time period. Therefore, the seepage levels on the logs represent the location of transient water seepage and may not indicate the static groundwater level. Groundwater levels encountered during drilling can be deceptive, because seepage into the boring can be blocked or slowed by the auger itself.

It should be noted that groundwater levels vary seasonally with rainfall and other factors. We anticipate that groundwater could be found in more permeable soil layers, and between the looser near-surface soil and the underlying denser soil.

The stratification lines on the logs represent the approximate boundaries between soil types at the exploration locations. The actual transition between soil types may be gradual, and subsurface conditions can vary

between exploration locations. The logs provide specific subsurface information only at the locations tested. If a transition in soil type occurred between samples in the borings, the depth of the transition was interpreted. The relative densities and moisture descriptions indicated on the test boring logs are interpretive descriptions based on the conditions observed during drilling.

CONCLUSIONS AND RECOMMENDATIONS

GENERAL

THIS SECTION CONTAINS A SUMMARY OF OUR STUDY AND FINDINGS FOR THE PURPOSES OF A GENERAL OVERVIEW ONLY. MORE SPECIFIC RECOMMENDATIONS AND CONCLUSIONS ARE CONTAINED IN THE REMAINDER OF THIS REPORT. ANY PARTY RELYING ON THIS REPORT SHOULD READ THE ENTIRE DOCUMENT.

The test borings conducted for this study encountered native, very stiff silt and harder silt, and dense silty sand, but there was a thick mantle of loose fill, and looser native soils approximately 15 to 25 feet thick overlying the dense soils. The high groundwater level makes these wet loose to medium dense soils subject to liquefaction during the design seismic event. Based on this information, we would not recommend relying on existing foundations to support the new home without foundation underpinning; and all new foundations should also be pile supported to transfer the loads to the underlying dense native soils which are not subject to settlement or liquefaction. Considering this, we recommend that the new structural loads from the proposed additions be supported on a deep foundation system consisting of small diameter pipe piles. The pipe piles would be driven through any looser soils, to refusal in the underlying competent, glacially compressed soils that comprise the core of the site. Settlement sensitive floors, decks, or other site features should be supported by these piles as well. Additional recommendations can be found in the ***Pipe Piles*** section of this report.

Projects involving small-diameter pipe piles often include the need for lateral resistance from fill placed against the foundations. If this is the case for this project, it is important that the structural engineer indicate this requirement on the plans for the general and earthwork contractor's information. Compaction requirements for this fill are discussed below in ***Pipe Piles***. The building department may require that we verify suitable compaction of this fill prior to completion of the project.

The retaining walls for the new driveway expansion would need to be relatively flexible MSE walls or pile supported concrete retaining walls. We can supply recommendations for these walls once the plans are more complete.

Excavations of the project will likely be relatively discrete and limited to small, localized excavations made for the installation of the pipe piles and construction of the pile supported grade beams. Based on the soils encountered in the borings, a temporary excavation inclination of no steeper than a 1:1 (Horizontal:Vertical) is appropriate for this project. Vertical excavations should not occur on, or near the shared property lines, or near any settlement sensitive structures. At this time, it appears that excavations for the new additions will be able to be maintained within the property boundaries without the need for excavation easements or shoring. Any cuts below the water table will likely need shoring. We can provide shoring recommendations as necessary and appropriate in the future.

The site is underlain generally by fine-grained silty sand and silt; both of which are essentially impervious. The composition of the native soils would act to essentially stop the downward percolation of stormwater, which can often be evidenced by a perched groundwater layer atop the underlying impermeable soils. Considering the soil qualities, and the onsite steep slope, we do not recommend that concentrated infiltration or dispersion systems be utilized at this property. Any attempt to infiltrate or disperse onsite stormwater would only act to cause adverse drainage impacts to the existing residence's basement as well as the adjacent structures and could cause adverse impacts to the southern steep slope and property line retaining walls.

While the site is underlain at depth by competent glacially compressed soils that are not susceptible to deep seated instability, the surficial soils on the slope are prone to further erosion over time. We recommend that all surface and subsurface water collected by the addition project be directed away from the top of the steep slope, and no additional fill soils be placed between the residence and top of the southern steep slope, as both poorly managed stormwater and fill placement are common triggers for slope instability in the Pacific Northwest.

The erosion control measures needed during the site development will depend heavily on the weather conditions that are encountered. We anticipate that a silt fence will be needed around the downslope sides of any cleared areas. Existing pavements, ground cover, and landscaping should be left in place wherever possible to minimize the amount of exposed soil. Rocked staging areas and construction access roads should be provided to reduce the amount of soil or mud carried off the property by trucks and equipment. Trucks should not be allowed to drive off of the rock-covered areas. Cut slopes and soil stockpiles should be covered with plastic during wet weather. Following clearing or rough grading, it may be necessary to mulch or hydroseed bare areas that will not be immediately covered with landscaping or an impervious surface. On most construction projects, it is necessary to periodically maintain or modify temporary erosion control measures to address specific site and weather conditions.

The drainage and/or waterproofing recommendations presented in this report are intended only to prevent active seepage from flowing through concrete walls or slabs. Even in the absence of active seepage into and beneath structures, water vapor can migrate through walls, slabs, and floors from the surrounding soil, and can even be transmitted from slabs and foundation walls due to the concrete curing process. Water vapor also results from occupant uses, such as cooking, cleaning, and bathing. Excessive water vapor trapped within structures can result in a variety of undesirable conditions, including, but not limited to, moisture problems with flooring systems, excessively moist air within occupied areas, and the growth of molds, fungi, and other biological organisms that may be harmful to the health of the occupants. The designer or architect must consider the potential vapor sources and likely occupant uses, and provide sufficient ventilation, either passive or mechanical, to prevent a build up of excessive water vapor within the planned structure.

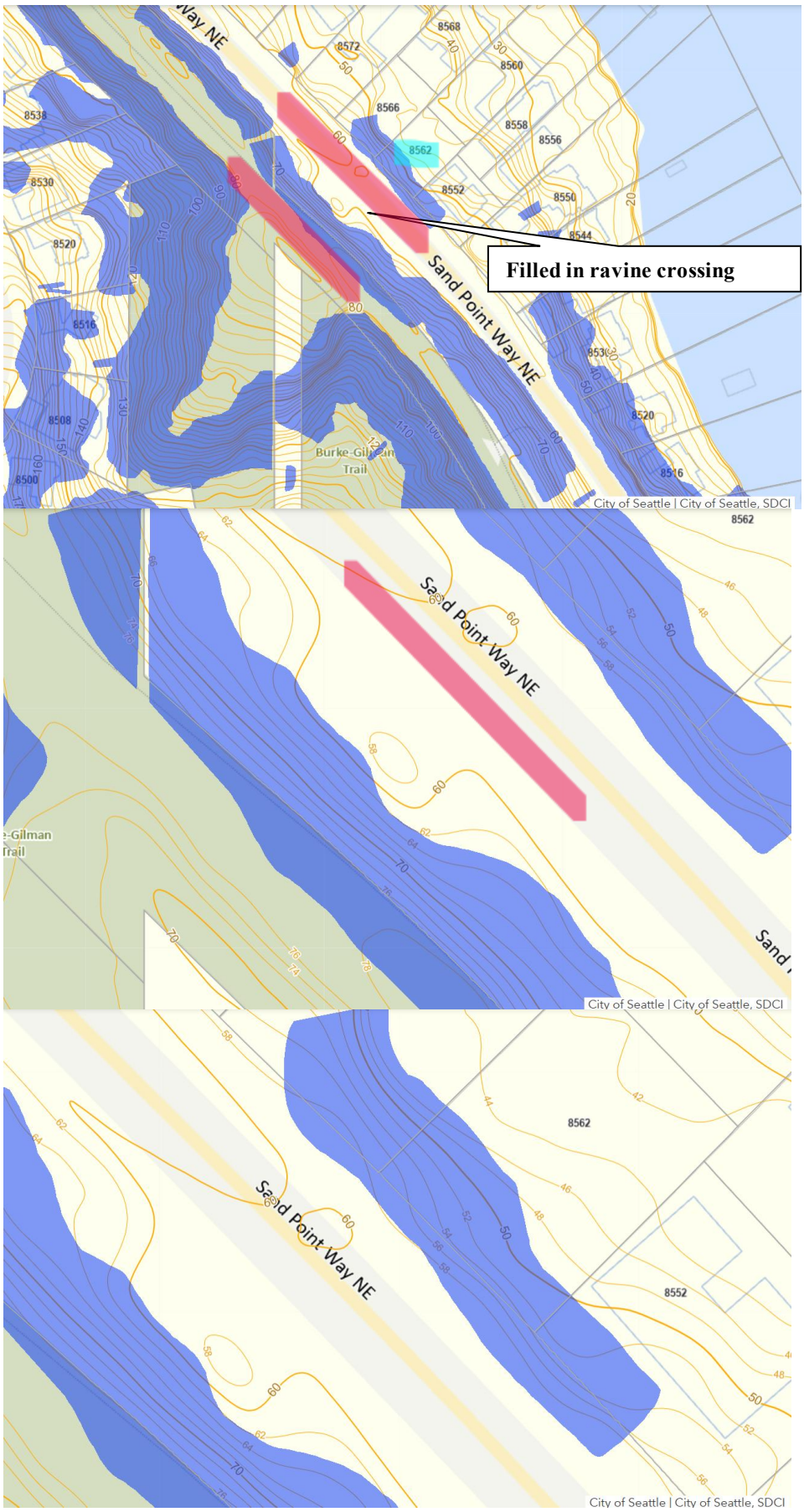
Steep Slope Erosion Hazard Area Relief

Any slope in Seattle that has an inclination of 40 percent or more and a height of 10 feet or more is considered a Steep Slope Erosion Hazard Area, per Seattle Municipal Code (SMC). A Steep Slope Erosion Hazard Area exists on the southwestern portion of the subject site. SMC 25.09.090.B.1 states that development on Steep Slope Erosion Hazard Areas is prohibited unless "relief" can be obtained. Based on SMC 25.09.090.B.2.b, relief from the prohibition of development can be obtained for Steep Slope Erosion Hazard Areas when it can be proven that the steep slope erosion hazard area has been modified through previous legal grading activities if no adverse impact on the steep slope erosion hazard area will result.

<u>We believe that relief from development restrictions of the Steep Slope Erosion Hazard Area on the site, as noted in SMC 25.09.090.B.2.b, is warranted based on the following reasons:</u>

1. Based on the three clips of the GIS mapping (below), the topography indicates that the steep slopes were created by filling in across a gently inclined ravine feature during construction of the Burke Gilman Rail and for Sand Point Way Northeast.
2. The addition and existing foundations will be supported on piles embedded into the competent underlying soils which are not subject to slope instability.
3. The only elements of the project within the buffers of the steep slopes involves filling for the driveway, which will stabilize the slopes in the area of the filling.

In summary, it is our professional opinion from a geotechnical engineering standpoint that, provided recommendations in this study are followed, gaining relief from the Steep Slope Erosion Hazard Areas development standards of the SMC is appropriate for this project.


Filled in ravine crossing
Sand Point Way NE
Burke-Gilman Trail
8562
8552
City of Seattle | City of Seattle, SDCI

Geotech Consultants, Inc. should be allowed to review the final development plans to verify that the recommendations presented in this report are adequately addressed in the design. Such a plan review would be additional work beyond the current scope of work for this study, and it may include revisions to our recommendations to accommodate site, development, and geotechnical constraints that become more evident during the review process.

We recommend including this report, in its entirety, in the project contract documents. This report should also be provided to any future property owners so they will be aware of our findings and recommendations.

SEISMIC CONSIDERATIONS

In accordance with the International Building Code (IBC), the site class within 100 feet of the ground surface is best represented by Site Class Type F (Failure-Prone Site Class). However, the code allows for an exception from the F classification if the building period is less than 0.5 seconds. We anticipate the proposed residence will be less than 0.5 seconds, and therefore a Site Class Type D can be used for the project. This will need to be confirmed by the project structural engineer. As noted in the USGS website, the mapped spectral acceleration value for a 0.2 second (S_s) and 1.0 second period (S_1) equals 1.27g and 0.44g, respectively.

The IBC and ASCE 7 require that the potential for liquefaction (soil strength loss) during an earthquake be evaluated for the peak ground acceleration of the Maximum Considered Earthquake (MCE), which has a probability of occurring once in 2,475 years (2 percent probability of occurring in a 50-year period). The MCE peak ground acceleration adjusted for site class effects (F_{PGA}) equals 0.59. The loose, near-surface, saturated soils beneath the site are susceptible to seismic liquefaction under the ground motions of the MCE. The recommendations presented in this report to support the new residence on deep foundations embedded into the dense underlying glacially compressed soils not susceptible to seismic liquefaction are intended to prevent catastrophic failure of the proposed new structures and equipment in the event of a large seismic event.

Sections 1803.5 of the IBC and 11.8 of ASCE 7 require that other seismic-related geotechnical design parameters (seismic surcharge for retaining wall design and slope stability) include the potential effects of the Design Earthquake. The peak ground acceleration for the Design Earthquake is defined in Section 11.2 of ASCE 7 as two-thirds (2/3) of the MCE peak ground acceleration, or 0.39.

Summary of Liquefaction Analysis

Based on the borings conducted at the site, liquefaction of the saturated soils below a depth of approximately 3 to 5 feet could occur during the design earthquake beneath the site. Current geotechnical analysis cannot accurately predict where, and to what extend soil liquefaction will occur during a large earthquake. Using procedures developed by Seed, Idress, et al., we have calculated the approximate total ground settlement that could result if liquefaction were to occur in the saturated, loose to medium-dense soils as a result of the design earthquake for this site, and for sites in the vicinity. Based on these analyses, it is probable that settlement in the liquefaction prone soil layers during the MCE could have a total ground settlement in the order of 4 inches. This liquefaction will predominantly occur within the loose to medium-dense saturated soils between the groundwater table (approximately 5 foot below grade), above the glacially compressed sands below, which have a low potential for liquefaction. Liquefaction calculations are attached to this report.

Lateral Spreading

The potential for lateral spreading in the event of a large earthquake is even less understood than liquefaction itself. However, some methods have been developed to estimate the potential amount of lateral ground movement that could occur where liquefiable sites are in close proximity to free face conditions, such as along the shoreline of Puget Sound. Our analysis using procedures developed by Bartlett and Youd (attached) showed negligible potential for lateral ground movement during the MCE.

PIPE PILES

A 2-inch-diameter pipe pile driven with a minimum 90-pound jackhammer or a 140-pound Rhino hammer to a final penetration rate of 1-inch or less for one minute of continuous driving may be assigned an allowable compressive load of 3 tons.

Three- and four-inch-diameter pipe piles driven with an 850- or 1,100- or 2,000-pound hydraulic jackhammer to the following final penetration rates may be assigned the following compressive capacities.

INSIDE PILE DIAMETER	FINAL DRIVING RATE (850-pound hammer)	FINAL DRIVING RATE (1,100-pound hammer)	FINAL DRIVING RATE (2,000-pound hammer)	ALLOWABLE COMPRESSIVE CAPACITY
3 inches	10 sec/inch	6 sec/inch	2 sec/inch	6 tons
4 inches	16 sec/inch	10 sec/inch	4 sec/inch	10 tons

Note: The refusal criteria indicated in the above table are valid only for pipe piles that are installed using a hydraulic impact hammer carried on leads that allow the hammer to sit on the top of the pile during driving. If the piles are installed by alternative methods, such as a vibratory hammer or a hammer that is hard-mounted to the installation machine, numerous load tests to 200 percent of the design capacity would be necessary to substantiate the allowable pile load. The appropriate number of load tests would need to be determined at the time the contractor and installation method are chosen.

Extra-strong, Schedule 80 steel pipe should be used for 2-inch-diameter piles. At a minimum, Schedule 40 pipe should be used for 3-inch-diameter or larger piles. The site soils are not highly organic, but the piles will be very near the groundwater surface. Considering this, it is our opinion that corrosion protection, such as galvanizing, is necessary for the pipe piles.

Seattle Director's Rule 10-2009 contains several prescriptive requirements related to the use of pipe piles having a diameter of less than 10 inches. Under Director's Rule 10-2009, load tests are not required for 2-inch-diameter piles that are designed for an allowable 3-ton capacity. Load tests and a code alternate or modification would be required if alternative installation methods are used, or if a higher capacity is desired. The City of Seattle limits the length of 2-inch-diameter pipe piles to 30 feet. If pile lengths exceed the 30 feet, a code alternate or modification must be applied for. Under Director's Rule 10-2009, load tests are required on 3 percent of the installed piles (3-inch-diameter or larger) up to a maximum of 5 piles, with a minimum of one pile load test on each project. Additionally, full-time observation of the pile installation by the geotechnical engineer-of-record is required by Director's Rule 10-2009.

We recommend a minimum pile length of 15 feet below the existing ground surface. Our experience with installation of small-diameter pipe piles indicates that it is likely that they will be longer than this minimum length to reach refusal.

Pile caps and grade beams should be used to transmit loads to the piles. Isolated pile caps should include a minimum of two piles to reduce the potential for eccentric loads being applied to the piles. Subsequent sections of pipe can be connected with slip or threaded couplers, or they can be welded together. If slip couplers are used, they should fit snugly into the pipe sections. This may require that shims be used or that beads of welding flux be applied to the outside of the coupler.

Lateral loads due to wind or seismic forces may be resisted by passive earth pressure acting on the vertical, embedded portions of the foundation. For this condition, the foundation must be either poured directly against relatively level, undisturbed soil or be surrounded by level compacted fill. We recommend using a passive earth pressure of 250 pounds per cubic foot (pcf) for this resistance. We recommend a safety factor of at least 1.5 for the foundation's resistance to lateral loading, when using the above ultimate passive value.

FOUNDATION AND RETAINING WALLS

Retaining walls backfilled on only one side should be designed to resist the lateral earth pressures imposed by the soil they retain. The following recommended parameters are for walls that restrain level backfill:

PARAMETER	VALUE
Active Earth Pressure *	40 pcf
Passive Earth Pressure	250 pcf
Soil Unit Weight	130 pcf

Where: pcf is Pounds per Cubic Foot, and Active and Passive Earth Pressures are computed using the Equivalent Fluid Pressures.
*** For a restrained wall that cannot deflect at least 0.002 times its height, a uniform lateral pressure equal to 10 psf times the height of the wall should be added to the above active equivalent fluid pressure.**

The design values given above do not include the effects of any hydrostatic pressures behind the walls and assume that no surcharges, such as those caused by slopes, vehicles, or adjacent foundations will be exerted on the walls. If these conditions exist, those pressures should be added to the above lateral soil pressures. Where sloping backfill is desired behind the walls, we will need to be given the wall dimensions and the slope of the backfill in order to provide the appropriate design earth pressures. The surcharge due to traffic loads behind a wall can typically be accounted for by adding a uniform pressure equal to 2 feet multiplied by the above active fluid density. Heavy construction equipment should not be operated behind retaining and foundation walls within a distance equal to the height of a wall, unless the walls are designed for the additional lateral pressures resulting from the equipment.

The values given above are to be used to design only permanent foundation and retaining walls that are to be backfilled, such as conventional walls constructed of reinforced concrete or masonry. It is not appropriate to use the above earth pressures and soil unit weight to back-calculate soil strength parameters for design of other types of retaining walls, such as soldier pile, reinforced earth, modular or soil nail walls. We can assist with design of these types of walls, if desired. The passive pressure given is appropriate only for a shear key poured directly against undisturbed native soil, or for the depth of level, well-compacted fill placed in front of a retaining or foundation wall. The values for friction and passive resistance are ultimate values and do not include a safety factor. Restrained wall soil parameters should be utilized for a distance of 1.5 times the wall height from corners or bends in the walls. This is intended to reduce the amount of cracking that can occur where a wall is restrained by a corner.

Wall Pressures Due to Seismic Forces

The surcharge wall loads that could be imposed by the design earthquake can be modeled by adding a uniform lateral pressure to the above-recommended active pressure. The recommended surcharge pressure is 9**H** pounds per square foot (psf), where **H** is the design retention height of the wall. Using this increased pressure, the safety factor against sliding and overturning can be reduced to 1.2 for the seismic analysis.

Retaining Wall Backfill and Waterproofing

Backfill placed behind retaining or foundation walls should be coarse, free-draining structural fill containing no organics. This backfill should contain no more than 5 percent silt or clay particles and have no gravel greater than 4 inches in diameter. The percentage of particles passing the No. 4 sieve should be between 25 and 70 percent. Unless free-draining material is used to backfill the retaining walls, a drainage composite similar to Miradrain 6000 should be placed against the backfilled

retaining walls. The drainage composites should be hydraulically connected to the foundation drain system. Free-draining backfill or gravel should be used for the entire width of the backfill where seepage is encountered. For increased protection, drainage composites should be placed along cut slope faces, and the walls should be backfilled entirely with free-draining soil.

The purpose of these backfill requirements is to ensure that the design criteria for a retaining wall are not exceeded because of a build-up of hydrostatic pressure behind the wall. Also, subsurface drainage systems are not intended to handle large volumes of water from surface runoff. The top 12 to 18 inches of the backfill should consist of a compacted, relatively impermeable soil or topsoil, or the surface should be paved. The ground surface must also slope away from backfilled walls to reduce the potential for surface water to percolate into the backfill. Water percolating through pervious surfaces (pavers, gravel, permeable pavement, etc.) must also be prevented from flowing toward walls or into the backfill zone. The compacted subgrade below pervious surfaces and any associated drainage layer should therefore be sloped away. Alternatively, a membrane and subsurface collection system could be provided below a pervious surface.

It is critical that the wall backfill be placed in lifts and be properly compacted, in order for the above-recommended design earth pressures to be appropriate. The wall design criteria assume that the backfill will be well-compacted in lifts no thicker than 12 inches. The compaction of backfill near the walls should be accomplished with hand-operated equipment to prevent the walls from being overloaded by the higher soil forces that occur during compaction. The section entitled ***General Earthwork and Structural Fill*** contains additional recommendations regarding the placement and compaction of structural fill behind retaining and foundation walls.

The above recommendations are not intended to waterproof below-grade walls, or to prevent the formation of mold, mildew or fungi in interior spaces. Over time, the performance of subsurface drainage systems can degrade, subsurface groundwater flow patterns can change, and utilities can break or develop leaks. Therefore, waterproofing should be provided where future seepage through the walls is not acceptable. This typically includes limiting cold-joints and wall penetrations, and using bentonite panels or membranes on the outside of the walls. There are a variety of different waterproofing materials and systems, which should be installed by an experienced contractor familiar with the anticipated construction and subsurface conditions. Applying a thin coat of asphalt emulsion to the outside face of a wall is not considered waterproofing, and will only help to reduce moisture generated from water vapor or capillary action from seeping through the concrete. As with any project, adequate ventilation of basement and crawl space areas is important to prevent a buildup of water vapor that is commonly transmitted through concrete walls from the surrounding soil, even when seepage is not present. This is appropriate even when waterproofing is applied to the outside of foundation and retaining walls. We recommend that you contact an experienced envelope consultant if detailed recommendations or specifications related to waterproofing design, or minimizing the potential for infestations of mold and mildew are desired.

BUILDING FLOORS

New settlement sensitive building floors should be supported by the pipe piles, either as a structural slab, or framed floor over a crawlspace. If some cracking an potential settlement is acceptable in the garage slabs, they may be supported on a minimum of 1 foot of new structural fill and reinforced with N0.4 rebar on 16 inches on center.

Even where the exposed soils appear dry, water vapor will tend to naturally migrate upward through the soil to the new constructed space above it. This can affect moisture-sensitive flooring, cause imperfections or damage to the slab, or simply allow excessive water vapor into the space above the slab. All interior slabs-on-grade should be underlain by a capillary break drainage layer consisting of a minimum 4-inch thickness of clean gravel or crushed rock that has a fines content (percent passing the No. 200 sieve) of less than 3 percent and a sand content (percent passing the No. 4 sieve) of no more than 10 percent. Pea gravel or crushed rock are typically used for this layer.

As noted by the American Concrete Institute (ACI) in the *Guides for Concrete Floor and Slab Structures*, proper moisture protection is desirable immediately below any on-grade slab that will be covered by tile, wood, carpet, impermeable floor coverings, or any moisture-sensitive equipment or products. ACI recommends a minimum 10-mil thickness vapor retarder for better durability and long term performance than is provided by 6-mil plastic sheeting that has historically been used. A vapor retarder is defined as a material with a permeance of less than 0.3 perms, as determined by ASTM E 96. It is possible that concrete admixtures may meet this specification, although the manufacturers of the admixtures should be consulted. Where vapor retarders are used under slabs, their edges should overlap by at least 6 inches and be sealed with adhesive tape. The sheeting should extend to the foundation walls for maximum vapor protection.

If no potential for vapor passage through the slab is desired, a vapor *barrier* should be used. A vapor barrier, as defined by ACI, is a product with a water transmission rate of 0.01 perms when tested in accordance with ASTM E 96. Reinforced membranes having sealed overlaps can meet this requirement.

We recommend that the contractor, the project materials engineer, and the owner discuss these issues and review recent ACI literature and ASTM E-1643 for installation guidelines and guidance on the use of the protection/blotter material.

EXCAVATIONS AND SLOPES

Temporary excavation slopes should not exceed the limits specified in local, state, and national government safety regulations. Also, temporary cuts should be planned to provide a minimum 2 to 3 feet of space for construction of foundations, walls, and drainage. Temporary cuts to a maximum overall depth of about 4 feet may be attempted vertically in unsaturated soil if there are no indications of slope instability. However, vertical cuts should not be made near property boundaries, or existing utilities and structures. Based upon Washington Administrative Code (WAC) 296, Part N, the soil at the subject site would generally be classified as Type B. Therefore, temporary cut slopes greater than 4 feet in height should not be excavated at an inclination steeper than 1:1 (Horizontal:Vertical), extending continuously between the top and the bottom of a cut. Any cuts below the water table will likely need shoring. Please contact Geotech if these cuts are planned.

The above-recommended temporary slope inclination is based on the conditions exposed in our explorations, and on what has been successful at other sites with similar soil conditions. It is possible that variations in soil and groundwater conditions will require modifications to the inclination at which temporary slopes can stand. Temporary cuts are those that will remain unsupported for a relatively short duration to allow for the construction of foundations, retaining walls, or utilities. Temporary cut slopes should be protected with plastic sheeting during wet weather. It is also important that surface runoff be directed away from the top of temporary slope cuts. Cut slopes should also be backfilled or retained as soon as possible to reduce the potential for instability. Please note that loose soil can cave suddenly and without warning. Excavation, foundation, and utility contractors should be made especially aware of this potential danger. These recommendations may need to be modified if the area near the potential cuts has been disturbed in the past by utility installation, or if settlement-sensitive utilities are located nearby.

All unretained permanent slopes at the site should be inclined no steeper than 2.5:1 (H:V).

Water should not be allowed to flow uncontrolled over the top of any temporary or permanent slope. All permanently exposed slopes should be seeded with an appropriate species of vegetation to reduce erosion and improve the stability of the surficial layer of soil.

Any disturbance to the existing slope outside of the proposed building limits may reduce the stability of the slope. Damage to the existing vegetation and ground should be minimized, and any disturbed areas should be revegetated as soon as possible. Soil from the excavation should not be placed on the slope, and this may require the off-site disposal of any surplus soil.

DRAINAGE CONSIDERATIONS

Footing drains should be used where: (1) crawl spaces or basements will be below a structure; (2) a slab is below the outside grade; or (3) the outside grade does not slope downward from a building. Drains should also be placed at the base of all earth-retaining walls. These drains should be surrounded by at least 6 inches of 1-inch-minus, washed rock that is encircled with non-woven, geotextile filter fabric (Mirafi 140N, Supac 4NP, or similar material). At its highest point, a perforated pipe invert should be at least 6 inches below the bottom of a slab floor or the level of a crawl space. The discharge pipe for subsurface drains should be sloped to flow to the outlet point. Roof and surface water drains must not discharge into the foundation drain system. A typical footing drain detail is attached to this report as Plate 5. For the best long-term performance, perforated PVC pipe is recommended for all subsurface drains. Clean-outs should be provided for potential future flushing or cleaning of footing drains.

As a minimum, a vapor retarder, as defined in the ***Building Floors*** section, should be provided in any crawl space area to limit the transmission of water vapor from the underlying soils. Crawl space grades are sometimes left near the elevation of the bottom of the footings. As a result, an outlet drain is recommended for all crawl spaces to prevent an accumulation of any water that may bypass the footing drains. Providing a few inches of free draining gravel underneath the vapor retarder is also prudent to limit the potential for seepage to build up on top of the vapor retarder.

Groundwater was observed during our field work well below the base of any excavation. However, if seepage is encountered in an excavation, it should be drained from the site by directing it through drainage ditches, perforated pipe, or French drains, or by pumping it from sumps interconnected by shallow connector trenches at the bottom of the excavation.

The excavation and site should be graded so that surface water is directed away from the tops of slopes. Water should not be allowed to stand in any area where foundations, slabs, or pavements are to be constructed. Final site grading in areas adjacent to the residence should slope away at least one to 2 percent, except where the area is paved. Surface drains should be provided where necessary to prevent ponding of water behind foundation or retaining walls. A discussion of grading and drainage related to pervious surfaces near walls and structures is contained in the ***Foundation and Retaining Walls*** section.

GENERAL EARTHWORK AND STRUCTURAL FILL

All building and pavement areas should be stripped of surface vegetation, topsoil, organic soil, and other deleterious material. It is important that existing foundations be removed before site development. The stripped or removed materials should not be mixed with any materials to be used as structural fill, but they could be used in non-structural areas, such as landscape beds.

Structural fill is defined as any fill, including utility backfill, placed under, or close to, a building, or in other areas where the underlying soil needs to support loads. All structural fill should be placed in horizontal lifts with a moisture content at, or near, the optimum moisture content. The optimum moisture content is that moisture content that results in the greatest compacted dry density. The moisture content of fill is very important and must be closely controlled during the filling and compaction process.

The allowable thickness of the fill lift will depend on the material type selected, the compaction equipment used, and the number of passes made to compact the lift. The loose lift thickness should not exceed 12 inches, but should be thinner if small, hand-operated compactors are used. We recommend testing structural fill as it is placed. If the fill is not sufficiently compacted, it should be recompacted before another lift is placed. This eliminates the need to remove the fill to achieve the required compaction. The following table presents recommended levels of relative compaction for compacted fill:

LOCATION OF FILL PLACEMENT	MINIMUM RELATIVE COMPACTION
Beneath walkways	95%
Filled slopes and behind retaining walls	90%
Beneath pavements	95% for upper 12 inches of subgrade; 90% below that level

Where: Minimum Relative Compaction is the ratio, expressed in percentages, of the compacted dry density to the maximum dry density, as determined in accordance with ASTM Test Designation D 1557-91 (Modified Proctor).

LIMITATIONS

The conclusions and recommendations contained in this report are based on site conditions as they existed at the time of our exploration and assume that the soil and groundwater conditions encountered in the test borings are representative of subsurface conditions on the site. If the subsurface conditions encountered during construction are significantly different from those observed in our explorations, we should be advised at once so that we can review these conditions and reconsider our recommendations where necessary. Unanticipated conditions are commonly encountered on construction sites and cannot be fully anticipated by merely taking samples in test borings. Subsurface conditions can also vary between exploration locations. Such unexpected conditions frequently require making additional expenditures to attain a properly constructed project. It is recommended that the owner consider providing a contingency fund to accommodate such potential extra costs and risks. This is a standard recommendation for all projects.

The recommendations presented in this report are directed toward the protection of only the proposed residence from damage due to slope movement. Predicting the future behavior of steep slopes and the potential effects of development on their stability is an inexact and imperfect science that is currently based mostly on the past behavior of slopes with similar characteristics. Landslides and soil movement can occur on steep slopes before, during, or after the development of property. The owner of any property containing or located close to steep slopes must ultimately accept the possibility that some slope movement could occur, resulting in possible loss of ground or damage to the facilities around the residence.

This report has been prepared for the exclusive use of Dave Richardson and his representatives, for specific application to this project and site. Our conclusions and recommendations are professional opinions derived in accordance with our understanding of current local standards of practice, and within the scope of our services. No warranty is expressed or implied. The scope of our services does not include services related to construction safety precautions, and our recommendations are not intended to direct the contractor's methods, techniques, sequences, or procedures, except as specifically described in our report for consideration in design. Our services also do not include assessing or minimizing the potential for biological hazards, such as mold, bacteria, mildew and fungi in either the existing or proposed site development.

ADDITIONAL SERVICES

In addition to reviewing the final plans, Geotech Consultants, Inc. should be retained to provide geotechnical consultation, testing, and observation services during construction. This is to confirm that subsurface conditions are consistent with those indicated by our exploration, to evaluate whether earthwork and foundation construction activities comply with the general intent of the recommendations presented in this report, and to provide suggestions for design changes in the event subsurface conditions differ from those anticipated prior to the start of construction. However, our work would not include the supervision or direction of the actual work of the contractor and its employees or agents. Also, job and site safety, and dimensional measurements, will be the responsibility of the contractor.

During the construction phase, we will provide geotechnical observation and testing services when requested by you or your representatives. Please be aware that we can only document sitework we actually observe. It

is still the responsibility of your contractor or on-site construction team to verify that our recommendations are being followed, whether we are present at the site or not.

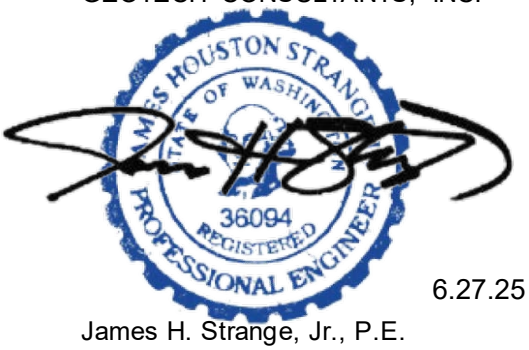
The following plates are attached to complete this report:

Plate 1	Vicinity Map
Plate 2	Site Exploration Plan
Plates 3 - 5	Test Boring Logs
Plate 6	Typical Footing Drain Detail
Attachments	Previous Borings and Liquefaction Calculations

We appreciate the opportunity to be of service on this project. Please contact us if you have any questions, or if we can be of further assistance.

Respectfully submitted,

GEOTECH CONSULTANTS, INC.

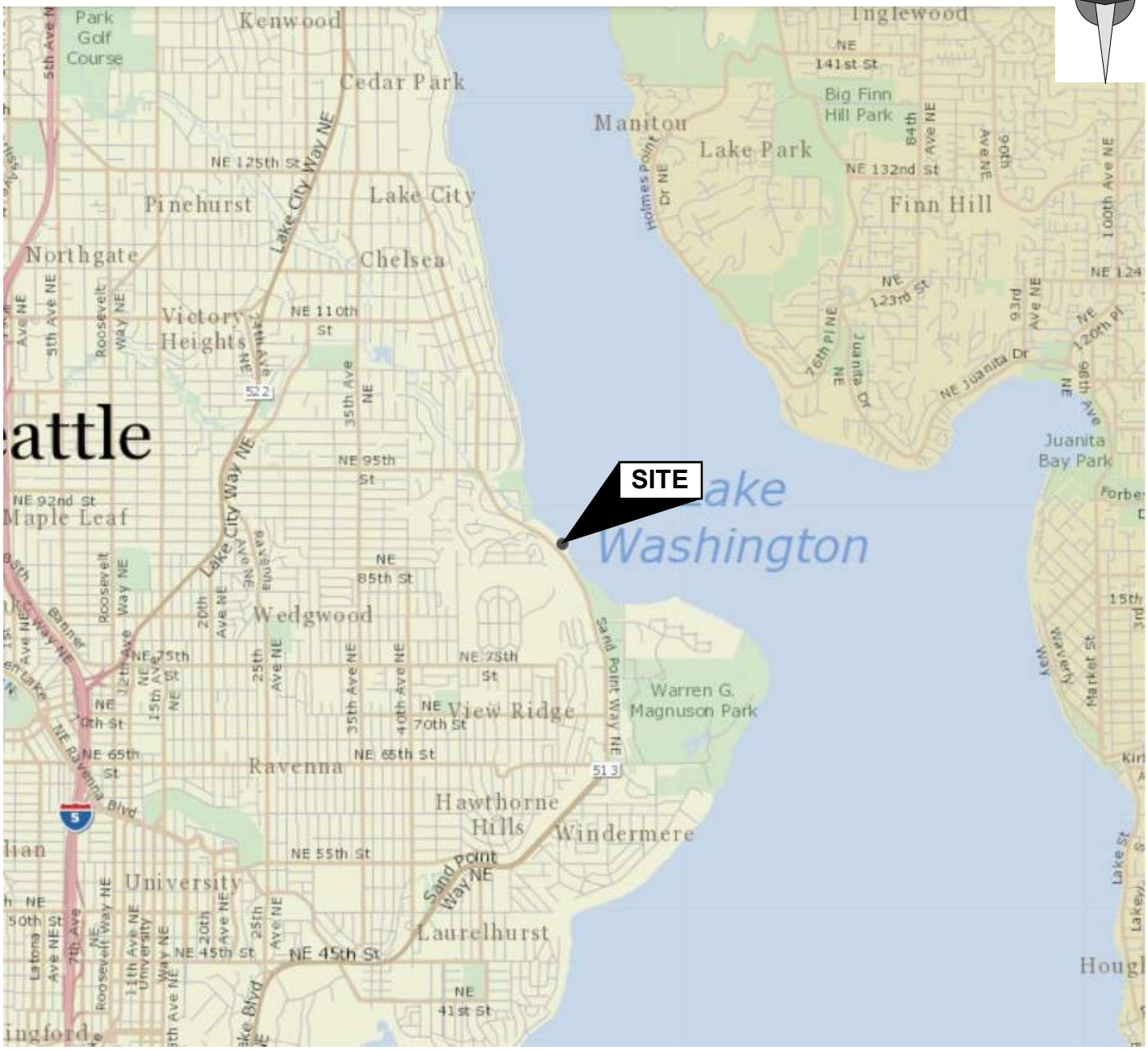


6.27.25

James H. Strange, Jr., P.E.
Associate

JHS:kg

NORTH



(Source: King County iMap)

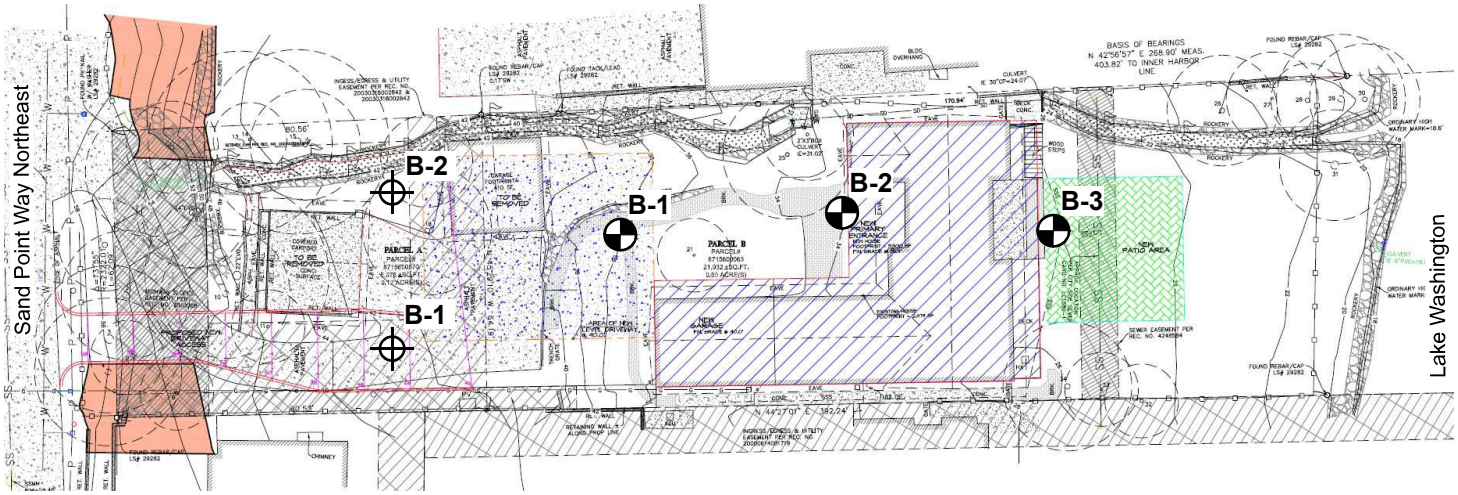
GEOTECH
CONSULTANTS, INC.

VICINITY MAP

8560 & 8562 Sand Point Way Northeast
Seattle, Washington

Job	Date:		Plate:
25173	June 2025		1

NORTH



Legend:

- Test Boring Location
- Test Boring Location (GCI - 2001)

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SITE EXPLORATION PLAN

8560 & 8562 Sand Point Way Northeast
Seattle, Washington

Job	Date:		Plate:
25173	June 2025	No Scale	2

BORING 1

Elev. ±36 feet

Depth (ft.)	Moisture	Water Table	Blows per Foot	Sample	USCS	Description
			6	1	SM	Gray silty SAND with gravel and occasional wood, fine to medium-grained, moist, loose - becomes wet
5			2	2	SM/ML	- grades to very silty SAND / sandy SILT with organics and decomposed wood, becomes very moist, loose
			16	3		- becomes medium-dense, with increased gravel, reduced organics, and decomposed wood
10			18	4	SP/SM	Gray slightly silty SAND with occasional organics and decomposed wood, fine to medium-grained, wet, medium-dense
15			49	5	SP/SM	Gray slightly silty SAND, fine-grained, moist, wet, dense
20			38	6	ML	Gray SILT, plastic, massive, moist, hard
25			50	7	SM	Greenish-blue-gray silty SAND with trace gravel, fine-grained, moist to very moist, very dense
30						

*** Test boring was terminated at 26.5 feet on June 4, 2025.**

*** Perched groundwater was encountered from 3 to 5 feet and groundwater was encountered below 9.5 feet during drilling.**

GEOTECH
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TEST BORING LOG

8560 & 8562 Sand Point Way Northeast
Seattle, Washington

Job	Date:	Logged by:	Plate:
25173	June 2025	ASM	3

BORING 2

Elev. ±34 feet

Moisture	Water Table	Blows per Foot	Sample	USCS	Description
				FILL	Brown SILT with sand, gravel, and occasional brick debris, fine to medium-grained, moist, loose **(FILL)**
		5	1	ML	Gray-brown SILT with rust-brown sand seams and occasional decomposed wood, non-plastic, moist, loose
	▽	11	2	SP	Gray SAND with gravel, silt, and decomposed wood, fine to medium-grained, wet, loose
		9	3	SM/ML	Gray very silty SAND / sandy SILT with gravel, occasional decomposed wood and organics, fine to medium-grained, wet, loose
		4	4		- grades to SILT with sand, occasional gravel, organics, and decomposed wood, becomes plastic, soft
		11	5	ML	- becomes greenish-blue-gray, stiff
		13	6		- with no gravel or decomposed wood
		51	7	SM/ML	Gray-brown with rust mottling silty SAND / sandy SILT, non-plastic, fine-grained, wet, very dense
		60	8	SM/SP	Greenish-blue-gray slightly silty SAND, fine-grained, wet, very dense
				SP	Gray SAND, fine-grained, wet, very dense

Depth scale: 5, 10, 15, 20, 25, 30, 35, 40

* **Test boring was terminated at 31.5 feet on June 4, 2025.**

* **Groundwater seepage was encountered below 5.5 feet during drilling.**

GEOTECH CONSULTANTS, INC.

TEST BORING LOG

8560 & 8562 Sand Point Way Northeast
Seattle, Washington

Job	Date:	Logged by:	Plate:
25173	June 2025	ASM	4

BORING 3

Elev. ±26 feet

Moisture	Water Table	Blows per Foot	Sample	USCS	Description
					Gray-brown with rust mottling, SILT with sand and occasional decomposed wood, plastic, moist, soft to medium-stiff
		6	1		
	▽	5	2		- becomes gray, wet, with occasional gravel, medium-stiff
		5	3		- becomes sandy
		10	4	ML	- with organics and reduced sand content
		11	5		- becomes clayey, with slightly reduced organcis and trace gravel - becomes stiff
		17	6		- becomes greenish-blue-gray, with trace organics and no gravel
		49	7	ML/SM	Greenish-blue-gray sandy SILT / very silty SAND, fine-grained, wet, dense to very dense
		45	8	SM	- grades to silty SAND

5 10 15 20 25 30 35 40

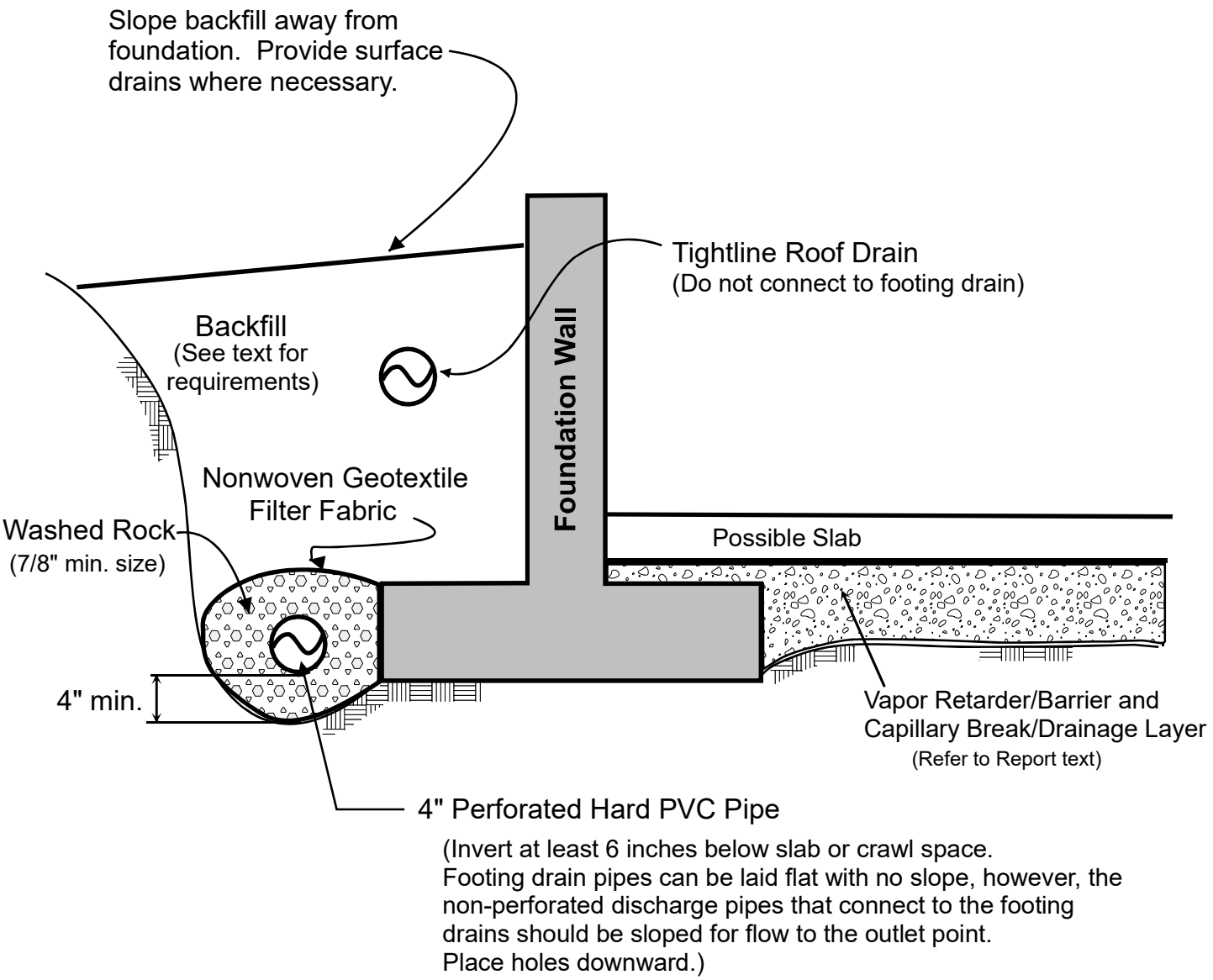
* **Test boring was terminated at 31.5 feet on June 4, 2025.**
* **Groundwater seepage was encountered below 5.5 feet during drilling.**

GEOTECH
CONSULTANTS, INC.

TEST BORING LOG

8560 & 8562 Sand Point Way Northeast
Seattle, Washington

Job	Date:	Logged by:	Plate:
25173	June 2025	ASM	5



NOTES:

(1) In crawl spaces, provide an outlet drain to prevent buildup of water that bypasses the perimeter footing drains.

(2) Refer to report text for additional drainage, waterproofing, and slab considerations.

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FOOTING DRAIN DETAIL

8560 & 8562 Sand Point Way Northeast
Seattle, Washington

Job	Date:		Plate:
25173	June 2025		6

BORING 1

Moisture Content	Water Table	Blows per Foot	Sample	USCS	Description
				FILL	Crushed rock base and gravel FILL
				ML	Brown and gray SILT with sand and gravel, very moist, loose -with organics -becomes wet
	▼	11	1	SM	Gray, silty, gravelly SAND, medium- to coarse-grained, wet, medium-dense
		15	2		-without gravel, becomes fine-to medium-grained, with interbedded, clayey lenses
		38	3		-becomes fine-grained, dense
		36	4		
		40	5		
		44	6		No sample recovered
		39	7		-becomes dilative

5 10 15 20 25 30 35 40

* **Test boring was terminated at 33 feet during drilling on September 6, 2001.**

* **Groundwater seepage was encountered at 5 feet during drilling.**

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BORING LOG

8560 Sandpoint Way Northeast
Seattle, Washington

Job No:	Date:	Logged by:	Plate:
01306	Oct. 2001	EMT	3

BORING 2

Moisture Content	Water Table	Blows per Foot	Sample	USCS	Description
					Dark brown, silty TOPSOIL, very moist, loose
	▼			SM	Dark brown, gravelly, silty SAND with organics, very moist, loose (**FILL**)
				SM	Gray, silty SAND, with trace gravel, fine- to medium-grained, wet, very loose
5		2	1		
10		7	2		-becomes slightly silty, SAND without gravel, fine- to medium-grained, wet, loose
					-becomes medium-dense
15		30	3		-becomes dense
20		26	4		
25		38	5	ML	Gray, slightly sandy SILT, non-plastic, slightly moist, hard
30					
35					
40					

* **Test boring was terminated at 26.5 feet during drilling on September 14, 2001.**

* **Groundwater seepage was encountered at 3 feet during drilling.**

GEOTECH
CONSULTANTS, INC.

BORING LOG

8560 Sandpoint Way Northeast
Seattle, Washington

Job No:	Date:	Logged by:	Plate:
01306	Oct. 2001	SES	4

JN 25173 - Richardson

Seismic Site Class Calculations

25173 - Seismic Class Calculation (B-1)			
Depth (ft)	N (blows/ft)	Layer Depth d (ft)	d/N
0-4.5	6	4.5	0.75
4.5-7	2	2.5	1.25
7-9.5	16	2.5	0.16
9.5-13	18	3.5	0.19
13-18	49	5	0.10
18-23	38	5	0.13
23-100	50	77	1.54
	Σ =	100	4.12

N = $\frac{\Sigma d}{\Sigma \frac{d}{N}}$ = **24** blows/ft

∴ Site Class D

25173 - Seismic Class Calculation (B-2)			
Depth (ft)	N (blows/ft)	Layer Depth d (ft)	d/N
0-4.5	5	4.5	0.90
4.5-7	11	2.5	0.23
7-9.5	9	2.5	0.28
9.5-13	4	3.5	0.88
13-18	11	5	0.45
18-23	13	5	0.38
23-28	51	5	0.10
28-100	60	72	1.20
	Σ =	100	4.55

N = $\frac{\Sigma d}{\Sigma \frac{d}{N}}$ = **22** blows/ft

∴ Site Class D

25173 - Seismic Class Calculation (B-3)			
Depth (ft)	N (blows/ft)	Layer Depth d (ft)	d/N
0-4.5	6	4.5	0.75
4.5-7	5	2.5	0.50
7-9.5	5	2.5	0.50
9.5-13	10	3.5	0.35
13-18	11	5	0.45
18-23	17	5	0.29
23-28	49	5	0.10
28-100	45	72	1.60
	Σ =	100	4.42

N = $\frac{\Sigma d}{\Sigma \frac{d}{N}}$ = **23** blows/ft

∴ Site Class D

ASCE 7-16

Table 20.3-1 Site Classification

Site Class	$\bar{v}_s$	$\bar{N}$ or $\bar{N}_{ch}$	$\bar{s}_u$
A. Hard rock	>5,000 ft/s	NA	NA
B. Rock	2,500 to 5,000 ft/s	NA	NA
C. Very dense soil and soft rock	1,200 to 2,500 ft/s	>50 blows/ft	>2,000 lb/ft²
D. Stiff soil	600 to 1,200 ft/s	15 to 50 blows/ft	1,000 to 2,000 lb/ft²
E. Soft clay soil	<600 ft/s	<15 blows/ft	<1,000 lb/ft²
	Any profile with more than 10 ft of soil that has the following characteristics: — Plasticity index $PI > 20$, — Moisture content $w \geq 40\%$, — Undrained shear strength $\bar{s}_u < 500$ lb/ft²		
F. Soils requiring site response analysis in accordance with Section 21.1	See Section 20.3.1		

Note: For SI: 1 ft = 0.3048 m; 1 ft/s = 0.3048 m/s; 1 lb/ft² = 0.0479 kN/m².

20.4 DEFINITIONS OF SITE CLASS PARAMETERS

The definitions presented in this section shall apply to the upper 100 ft (30 m) of the site profile. Profiles containing distinct soil and rock layers shall be subdivided into those layers designated by a number that ranges from 1 to n at the bottom where there are a total of n distinct layers in the upper 100 ft (30 m). Where some of the n layers are cohesive and others are not, k is the number of cohesive layers and m is the number of cohesionless layers. The symbol i refers to any one of the layers between 1 and n.

20.4.1 $\bar{v}_s$, Average Shear Wave Velocity. $\bar{v}_s$ shall be determined in accordance with the following formula:

$$\bar{v}_s = \frac{\sum_{i=1}^{n} d_i}{\sum_{i=1}^{n} \frac{d_i}{v_{si}}} \qquad (20.4\text{-}1)$$

where

d_i = the thickness of any layer between 0 and 100 ft (30 m);
v_{si} = the shear wave velocity in ft/s (m/s); and
$\sum_{i=1}^{n} d_i = 100$ ft (30 m).

20.4.2 $\bar{N}$, Average Field Standard Penetration Resistance and $\bar{N}_{ch}$, Average Standard Penetration Resistance for Cohesionless Soil Layers. $\bar{N}$ and $\bar{N}_{ch}$ shall be determined in accordance with the following formulas:

$$\bar{N} = \frac{\sum_{i=1}^{n} d_i}{\sum_{i=1}^{n} \frac{d_i}{N_i}} \qquad (20.4\text{-}2)$$

where N_i and d_i in Eq. (20.4-2) are for cohesionless soil, cohesive soil, and rock layers.

$$\bar{N}_{ch} = \frac{d_s}{\sum_{i=1}^{m} \frac{d_i}{N_i}} \qquad (20.4\text{-}3)$$

where N_i and d_i in Eq. (20.4-3) are for cohesionless soil layers only and

$$\sum_{i=1}^{m} d_i = d_s$$

where

d_s is the total thickness of cohesionless soil layers in the top 100 ft (30 m).
N_i is the standard penetration resistance (ASTM D1586) not to exceed 100 blows/ft (305 blows/m) as directly measured in the field without corrections.

Where refusal is met for a rock layer, N_i shall be taken as 100 blows/ft (305 blows/m).

20.4.3 $\bar{s}_u$, Average Undrained Shear Strength. $\bar{s}_u$ shall be determined in accordance with the following formula:

$$\bar{s}_u = \frac{d_c}{\sum_{i=1}^{k} \frac{d_i}{s_{ui}}} \qquad (20.4\text{-}4)$$

where

$\sum_{i=1}^{k} d_i = d_c$;
d_c = the total thickness of cohesive soil layers in the top 100 ft (30 m);
PI = the plasticity index as determined in accordance with ASTM D4318;
w = the moisture content in percent as determined in accordance with ASTM D2216; and
s_{ui} = the undrained shear strength in psf (kPa), not to exceed 5,000 psf (240 kPa) as determined in accordance with ASTM D2166 or ASTM D2850.

20.5 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

See Chapter 23 for the list of consensus standards and other documents that shall be considered part of this standard to the extent referenced in this chapter.

LIQUEFACTION AND SEISMIC SETTLEMENT ANALYSIS (SPT 2004)

Job No.:	25173	Boring No.:	B-2	Earthquake Magnitude, M_W:	7.14	Energy Ratio of Hammer (%):	60
Job Name:	Richardson	Ground Elevation* (ft):	100	Max. Ground Acc., a_{max} (g):	0.594	Sampling Method Corr., C_S:	1.2
Date:	6/27/2025	Water Elevation (ft):	95	Borehole Diameter (in):	5.9	Borehole Dia. Correction, C_B:	1.05

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
Sample No.	Sample Depth (ft)	Layer Bottom (ft)	γ (pcf)	r_d	σ_v (psf)	σ_v' (psf)	σ_v' (tsf)	C_N	C_R	N	$(N_1)_{60}$	FC (%)	$(N_1)_{60CS}$	K_σ	$CRR_{7.5}$	$CSR_{7.5}$	F.S.	$N_{Corr.}$	STRAIN CHART Vol. Strain (%)	Est. Settle. (in)
-	-	0	110																	
1	2.5	4.5	115	1.000	518	518	0.26	1.70	0.75	5	8	90	14	1.00	0.144	0.351	A.W.	14	2.1	-
2	5	7	125	0.993	830	830	0.42	1.45	0.80	11	16	50	22	1.00	0.229	0.349	0.7	20	1.5	0.5
3	7.5	9.5	125	0.985	1143	987	0.49	1.38	0.80	9	13	60	18	1.00	0.185	0.400	0.5	17	1.8	0.5
4	10	13	125	0.976	1580	1268	0.63	1.27	0.85	4	5	90	11	1.00	0.125	0.427	0.3	10	2.5	1.1
5	15	18	125	0.956	2205	1581	0.79	1.11	0.95	11	15	90	20	1.00	0.207	0.468	0.4	20	1.7	1.0
6	20	23	125	0.935	2830	1894	0.95	1.02	0.95	13	16	90	21	1.00	0.225	0.490	0.5	21	1.5	0.9
7	25	28	135	0.911	3505	2257	1.13	0.98	0.95	51	60	60	65	0.96	N.L.	0.515	N.L.	65	0	0.0
8	30	50	135	0.886	6475	4915	2.46	0.89	1.00	60	67	20	71	0.73	N.L.	0.558	N.L.	71	0	0.0

Total Saturated Settlement (in): 4.0

Notes: A.W. = Above Water

N.L. = Non-Liquefiable

* Ground surface elevation based on assumed datum of 100 feet.

A Sample number
B Avg. depth of blowcount interval below ground surface in feet
C Top of sample layer depth below ground surface in feet
D Total unit weight of soil in lb/ft^3
E Stress reduction factor, see Eqn. (6) (Idriss/Boulanger, 2004)
F Total vertical stress = γ_T * depth
G Effective vertical stress in lbs/ft^2 = σ_v - μ = Total - pore pressure
H Effective vertical stress in $tons/ft^2$
I Correction for effective overburden, see Eqn.(21) (Idriss/Boulanger, 2004)
J Correction for rod length, see Table 2 (Youd/Idriss, 2001)
K Standard penetration test blow count
L Normalized penetration resistance for overburden, see Eqn. (8) (Youd/Idriss, 2001)
M Percentage of soil sample passing #200 sieve, must be >0
N Adjusted N for % fines (FC), see Eqn. (25) (Idriss/Boulanger, 2004)
O Typical reduction in CRR with increase in initial confining pressure, see Eqn. 12 (Idriss/Boulanger, 2004)
P $CRR_{7.5}$ corrected for overburden, see Eqn(27) (Idriss/Boulanger, 2004)
Q Avg. CSR due to cylic loading scaled to M=7.5, see Eqn(23) (Idriss/Boulanger, 2004)
R Factor of Safety = (P) / (Q)
S Blow count possibly corrected for FC for settle. analysis (Martin/Lew, 1999)
T Volumetric Strain (%), estimate from STRAIN CHART
U Estimated Settlement by layer in inches

PREDICTION OF LATERAL SPREAD DISPLACEMENT

Job No.: 25173
Job Name: Richardson
Boring: B-2

FREE-FACE CONDITIONS (Eqn. 6a):

$$\log D_H = -16.713 + 1.532\,M - 1.406 \log R^* - 0.012\,R + 0.592 \log W + 0.540 \log T_{15} + 3.413 \log(100-F_{15}) - 0.795 \log(D50_{15}+0.1\text{mm})$$

$$\boldsymbol{D_H = 0.00}$$

GENTLY SLOPING GROUND CONDITIONS (Eqn. 6b):

$$\log D_H = -16.213 + 1.532\,M - 1.406 \log R^* - 0.012\,R + 0.338 \log S + 0.540 \log T_{15} + 3.413 \log(100-F_{15}) - 0.795 \log(D50_{15}+0.1\text{mm})$$

$$\boldsymbol{D_H = 0.00}$$

TERM	VALUE	DEFINITION	NOTES
D_H	**0.00**	the estimated lateral ground displacement, in meters (greater value of free-face or sloped conditions)	
M	7.14	the moment magnitude of the earthquake	$6 < M_W < 8$
R	56.0	the nearest horizontal or map distance from the site to the seismic energy source, in kilometers	5 km minimum
R^*	61.18	the modified source distance, in kilometers	(see PARAMETERS sheet)
W	4.6	the free-face ratio defined as the height (H) of the free face divided by the distance (L) from the base of the	$1 < W(\%) < 20$
		free face to the point in question, in percent	
S	0.1	the ground slope, in percent	$0.1 < S(\%) < 6$
T_{15}	5.48	the cumulative thickness of saturated granular layers with corrected blow counts, $(N_1)_{60}$, less than 15, in meters	$1 < T_{15}$ (m) < 15
F_{15}	76.0	the average fines content for granular materials included within T_{15}, in percent	Must be within bounds of
$D50_{15}$	0.15	the average mean grain size for the granular materials in T_{15}, in millimeters	Fig. 5 (Youd, et al, 2002)

Source: Youd, T. Leslie, Corbett M. Hansen, and Steven F. Bartlett, "Revised Multilinear Regression Equations for Prediction of Lateral Spread Displacement", *Journal of Geotechnical and Geoenvironmental Engineering*, Vol. 128, No. 12, December 1, 2002